

Applications

Special Bases

Base "e"

"e" is an irrational number (i.e. non-repeating, non-terminating) Its value is below.

$$e = 2.71828182845...$$

"e" is special because it is used in many applications, including finance and science.

Because base "e" powers are used so frequently, base "e" logarithms are abbreviated as follows:

$$\log_e X = \ln X$$

"ln" is short for "natural logarithm." Base "e" logarithms are given this name because they are used in real-world applications (e.g. biology, chemistry, physics, finance, etc.).

More Examples

$$\log_e 2 = \ln 2$$

$$\log_e \left(\frac{8}{7}\right) = \ln \left(\frac{8}{7}\right)$$

Base Ten

Base ten powers are special because it is easy to perform calculations with them. Consequently, they are used in many applications, including science (e.g. measuring the loudness of sound).

Because base ten powers are used so frequently, base ten logarithms are abbreviated as follows:

$$\log_{10} X = \log X$$

More Examples

$$\log_{10} 3 = \log 3$$

$$\log_{10} 8.2 = \log 8.2$$

Name
Date
Course

The Exponent Rules

Rules Learned in Pre-Algebra

$$\text{I) } x^a \cdot x^b = x^{a+b} \quad \text{Example: } x^2 \cdot x^5 = x^{2+5} = x^7$$

$$\text{II) } \frac{x^a}{x^b} = x^{a-b}, \quad x \neq 0 \quad \text{Example: } \frac{x^9}{x^3} = x^{9-3} = x^6$$

Rules Learned in Intermediate Algebra

$$\text{III) } (xy)^a = x^a \cdot y^a \quad \text{Example: } (2m)^4 = 2^4 \cdot m^4$$

$$\text{IV) } \left(\frac{x}{y}\right)^a = \frac{x^a}{y^a}, \quad y \neq 0 \quad \text{Example: } \left(\frac{n}{3}\right)^7 = \frac{n^7}{3^7}$$

$$\text{V) } (x^a)^b = x^{ab} \quad \text{Example: } (x^{10})^3 = x^{10 \cdot 3} = x^{30}$$

Strange Rules Learned in Intermediate Algebra

$$\text{VI) } x^1 = x \quad \text{Example: } 3^1 = 3$$

$$\text{VII) } x^0 = 1, \quad x \neq 0 \quad \text{Example: } 5^0 = 1$$

$$\text{VIII) } x^{-a} = \frac{1}{x^a}, \quad x \neq 0 \quad \text{Example: } x^{-2} = \frac{1}{x^2}$$

$$\text{IX) } x^{\frac{1}{a}} = \sqrt[a]{x} \quad \text{Example: } x^{\frac{1}{3}} = \sqrt[3]{x}$$

$$\text{X) } x^{\frac{b}{a}} = (\sqrt[a]{x})^b \quad \text{Example: } x^{\frac{5}{7}} = (\sqrt[7]{x})^5 \text{ or } \sqrt[7]{x^5}$$

Solving Exponential Equations with Special Bases

Solve the following equations. Write the Roman numeral of each exponent rule that you use in the margin. Round to the nearest thousandth if necessary.

① $7 = e^x \cdot e^{3x}$

$$7 = e^{x+3x}$$

$$7 = e^{4x}$$

$$4x = \log_e 7$$

$$4x = \ln 7$$

$$x = \frac{\ln 7}{4}$$

$$\boxed{x \approx 0.486}$$

II | ② $\frac{10^{7x}}{10^{5x}} = 8$

$$10^{7x-5x} = 8$$

$$10^{2x} = 8$$

$$2x = \log_{10} 8$$

$$2x = \log 8$$

$$x = \frac{\log 8}{2}$$

$$\boxed{x \approx 0.452}$$

• ③ $\frac{e^{12x}}{e^{7x}} = 3$

II $e^{12x-7x} = 3$

$$e^{5x} = 3$$

$$5x = \log_e 3$$

$$5x = \ln 3$$

$$x = \frac{\ln 3}{5}$$

$$\boxed{x \approx 0.22}$$

I | ④ $10^x \cdot 10^{2x} = 18$

$$10^{x+2x} = 18$$

$$10^{3x} = 18$$

$$3x = \log_{10} 18$$

$$3x = \log 18$$

$$x = \frac{\log 18}{3}$$

$$\boxed{x \approx 0.418}$$

• ⑤ $-8 = 1 - 2e^x$

$$-9 = -2e^x$$

$$\frac{-9}{-2} = \frac{-2e^x}{-2}$$

$$\frac{9}{2} = e^x$$

$$x = \log_e\left(\frac{9}{2}\right)$$

$$x = \ln\left(\frac{9}{2}\right)$$

$$\boxed{x \approx 1.504}$$

⑥ $6(10^x) + 4 = 16$

$$6(10^x) = 12$$

$$\frac{6(10^x)}{6} = \frac{12}{6}$$

$$10^x = 2$$

$$x = \log_{10} 2$$

$$x = \log 2$$

$$\boxed{x \approx 0.301}$$

• ⑦ $3e^x - 1 = 11$

$$3e^x = 12$$

$$\frac{3e^x}{3} = \frac{12}{3}$$

$$e^x = 4$$

$$x = \log_e 4$$

$$x = \ln 4$$

$$\boxed{x \approx 1.386}$$

⑧ $14 = 7 + 5(10^x)$

$$7 = 5(10^x)$$

$$\frac{7}{5} = \frac{5(10^x)}{5}$$

$$\frac{7}{5} = 10^x$$

$$x = \log_{10}\left(\frac{7}{5}\right)$$

$$x = \log\left(\frac{7}{5}\right)$$

$$\boxed{x \approx 0.146}$$

Solving Logarithmic Equations with Special Bases

Solve the following equations. Write the Roman numeral of each exponent rule that you use in the Margin.
Round to the nearest hundredth if necessary.

• ⑨ $2\log X - 5 = 5$

$$2\log_{10} X - 5 = 5$$

$$2\log_{10} X = 10$$

$$\frac{2\log_{10} X}{2} = \frac{10}{2}$$

$$\log_{10} X = 5$$

$$X = 10^5$$

$$\boxed{X = 100,000}$$

⑩ $13 = -\frac{\ln X}{3} + 15$

$$13 = -\frac{\log_e X}{3} + 15$$

$$-2 = -\frac{\log_e X}{3}$$

$$(-3)(-2) = -\frac{\log_e X}{3}(-3)$$

$$6 = \log_e X$$

$$X = e^6$$

$$\boxed{X \approx 403.43}$$

• ⑪ $-17 = 4\ln X - 21$

$$-17 = 4\log_e X - 21$$

$$4 = 4\log_e X$$

$$\frac{4}{4} = \frac{4\log_e X}{4}$$

$$1 = \log_e X$$

$$X = e^1$$

VI

$$X = e$$

$$\boxed{X \approx 2.72}$$

⑫ $\frac{\log X}{5} - 1 = -1$

$$\frac{\log_{10} X}{5} - 1 = -1$$

$$\frac{\log_{10} X}{5} = 0$$

$$(5)\frac{\log_{10} X}{5} = 0(5)$$

$$\log_{10} X = 0$$

$$X = 10^0$$

VII

$$\boxed{X = 1}$$

$$\log_b b^x = x$$

Loudness

The Loudness of Sound: The loudness $L(x)$ of sound in decibel units with intensity "x" measured in Watts per square meter is defined with the following equation.

$$L(x) = 10 \log(10^{12}x)$$

Given the following intensities of sound, find the loudness perceived by the human ear.

• (13) 10^{-8} W/m^2

$$L(10^{-8}) = 10 \log_{10}(10^{12}(10^{-8})) \stackrel{I}{=} 10 \log_{10}(10^4) = 10(4) = \boxed{40 \text{ dB}}$$

(14) 10^{-10} W/m^2

$$L(10^{-10}) = 10 \log_{10}(10^{12}(10^{-10})) \stackrel{I}{=} 10 \log_{10}(10^2) = 10(2) = \boxed{20 \text{ dB}}$$

(15) 10^{-14} W/m^2

$$L(10^{-14}) = 10 \log_{10}(10^{12}(10^{-14})) \stackrel{I}{=} 10 \log_{10}(10^{-2}) = 10(-2) = \boxed{-20 \text{ dB}}$$

• (16) 10^{-12} W/m^2

$$L(10^{-12}) = 10 \log_{10}(10^{12}(10^{-12})) \stackrel{I}{=} 10 \log_{10} 10^0 = 10(0) = \boxed{0 \text{ dB}}$$

(17) 10^{-9} W/m^2

$$L(10^{-9}) = 10 \log_{10}(10^{12}(10^{-9})) \stackrel{I}{=} 10 \log_{10}(10^3) = 10(3) = \boxed{30 \text{ dB}}$$

(18) 10^{-11} W/m^2

$$L(10^{-11}) = 10 \log_{10}(10^{12}(10^{-11})) \stackrel{I}{=} 10 \log_{10}(10^1) = 10(1) = \boxed{10 \text{ dB}}$$

Find the intensity of sound required to produce the following decibels.

• (19) 24 dB

$$L(x) = 10 \log(10^{12}x)$$

$$24 = 10 \log(10^{12}x)$$

$$\frac{24}{10} = \frac{10 \log(10^{12}x)}{10}$$

$$2.4 = \log(10^{12}x)$$

$$2.4 = \log_{10}(10^{12}x)$$

$$10^{12}x = 10^{2.4}$$

$$10^{12}x = 10^{2.4}$$

$$\frac{10^{12}x}{10^{12}} = \frac{10^{2.4}}{10^{12}}$$

$$x = 10^{-9.6} \text{ W/m}^2$$

②① 51 dB

$$L(x) = 10 \log(10^{12}x)$$

$$L(x) = 10 \log_{10}(10^{12}x)$$

$$51 = 10 \log_{10}(10^{12}x)$$

$$5.1 = \log_{10}(10^{12}x)$$

$$10^{12}x = 10^{5.1}$$

$$\frac{10^{12}x}{10^{12}} = \frac{10^{5.1}}{10^{12}}$$

II

$$X = 10^{-6.9} \text{ W/m}^2$$

②① 130 dB

$$L(x) = 10 \log(10^{12}x)$$

$$L(x) = 10 \log_{10}(10^{12}x)$$

$$130 = 10 \log_{10}(10^{12}x)$$

$$13 = \log_{10}(10^{12}x)$$

$$10^{12}x = 10^{13}$$

$$\frac{10^{12}x}{10^{12}} = \frac{10^{13}}{10^{12}}$$

II

$$x = 10^1$$

VI

$$X = 10 \text{ W/m}^2$$

• ②② 111 dB

$$L(x) = 10 \log(10^{12}x)$$

$$L(x) = 10 \log_{10}(10^{12}x)$$

$$111 = 10 \log_{10}(10^{12}x)$$

$$11.1 = \log_{10}(10^{12}x)$$

$$10^{12}x = 10^{11.1}$$

$$\frac{10^{12}x}{10^{12}} = \frac{10^{11.1}}{10^{12}}$$

II

$$x = 10^{-0.9}$$

$$X \approx 0.12589 \text{ W/m}^2$$

②③ 90 dB

$$L(x) = 10 \log(10^{12}x)$$

$$L(x) = 10 \log_{10}(10^{12}x)$$

$$90 = 10 \log_{10}(10^{12}x)$$

$$9 = \log_{10}(10^{12}x)$$

$$10^{12}x = 10^9$$

$$\frac{10^{12}x}{10^{12}} = \frac{10^9}{10^{12}}$$

II

$$x = 10^{-3}$$

$$x = \frac{1}{10^3}$$

$$x = \frac{1}{1000}$$

VIII

$$X = 0.001 \text{ W/m}^2$$

②④ 12 dB

$$L(x) = 10 \log(10^{12}x)$$

$$L(x) = 10 \log_{10}(10^{12}x)$$

$$12 = 10 \log_{10}(10^{12}x)$$

$$1.2 = \log_{10}(10^{12}x)$$

$$10^{12}x = 10^{1.2}$$

$$\frac{10^{12}x}{10^{12}} = \frac{10^{1.2}}{10^{12}}$$

$$x = 10^{-10.8} \text{ W/m}^2$$

II

Simple Interest Formula

$$A = P + Prt$$

A = Amount of money possessed after t years. P = Initial amount of money (i.e. principal), r = Interest rate (in decimal form).

Compound Interest Formula

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

A = Amount of money possessed after t years. P = Initial amount of money (i.e. principal), r = Yearly interest rate, also called annual percentage rate (APR). n = Number of times compounded per year.

- ②⑤ If \$7,592 is deposited in a bank account paying a 3% annual interest rate compounded monthly, how much money will be in the account in 2 years? Assume that there are no withdrawals or additional deposits.

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

$$A = 7,592\left(1 + \frac{0.03}{12}\right)^{12(2)}$$

$$A = 7,592(1.0025)^{24}$$

$$A = 7,592(1.0025)^{24}$$

$$A = 7,592(1.06175...)$$

$$A \approx \$8,060.86$$

- (26) If \$200.98 is deposited in a bank account paying a 0.8% annual interest rate compounded quarterly, how much money will be in the account in 5 years? Assume that there are no withdrawals or additional deposits

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

$$A = 200.98\left(1 + \frac{0.008}{4}\right)^{4(5)}$$

$$A = 200.98(1 + 0.002)^{20}$$

$$A = 200.98(1.002)^{20}$$

$$A = 200.98(1.0407...)$$

$$A \approx \boxed{\$209.17}$$

Continuous Compounding

$$A = P\left(1 + \frac{r}{n}\right)^{nt} = P\left[\left(1 + \frac{r}{n}\right)^n\right]^t = P\left[\left(1 + \frac{1}{\frac{n}{r}}\right)^n\right]^t = P\left[\left\{\left(1 + \frac{1}{\frac{n}{r}}\right)\right\}^{\frac{n}{r}}\right]^t$$

Exponent Rule V

Let n approach infinity.
As a result,
 $\left(1 + \frac{1}{\frac{n}{r}}\right)^{\frac{n}{r}}$

will approach 2.71828...

Therefore, if n approaches infinity, then

$$A = P\left[\{2.71828...\}^{rt}\right]^t$$

Exponent Rule V

$$A = P\{2.71828...\}^{rt}$$

We give this number a special letter designation.

$$A = Pe^{rt}$$

Continuous Compounding Formula

If n becomes very large (i.e. approaches infinity) in the compound interest formula, then the equation converts to the form below, where e is Euler's number (i.e. $\sim 2.718...$).

$$A = Pe^{rt}$$

- (27) If interest is compounded continuously for 2 years at a 3% rate, how much would a \$4,000 investment grow to?

$$A = Pe^{rt} = 4000e^{0.03(2)} = 4000e^{0.06} = 4000(1.0618...) = \boxed{\$4,247.35}$$

- (28) If a 0.07% interest rate is compounded continuously for 10 years, and the initial investment is \$541, how much would the investment be worth?

$$A = Pe^{rt} = 541e^{0.0007(10)} = 541e^{0.007} = 541(1.007...) = \boxed{\$544.80}$$

- (29) If a 1% interest rate is compounded continuously for 8 years, and the initial investment is \$947, how much would the investment be worth?

$$A = Pe^{rt} = 947e^{0.01(8)} = 947e^{0.08} = \boxed{\$1025.87}$$

- (30) If a 4.23% interest rate is compounded continuously for 15 years, and the initial investment is \$7,046, how much would the investment be worth?

$$A = Pe^{rt} = 7,046e^{0.0423(15)} = 7,046e^{0.6345} = 7,046(1.886...) = \boxed{\$13,289.31}$$

- (31) If an \$18,293 investment grows to \$40,368 in 9 years, what interest rate was applied if it was compounded continuously?

$$A = Pe^{rt}$$

$$40,368 = 18,293e^{r(9)}$$

$$40,368 = 18,293e^{9r}$$

$$\frac{40,368}{18,293} = \frac{18,293e^{9r}}{18,293}$$

$$2.2067... = e^{9r}$$

$$9r = \log_e 2.2067...$$

$$9r = \ln 2.2067...$$

$$9r = 0.7915...$$

$$r = 0.087946...$$

$$\boxed{8.79\%}$$

- (32) If an investment grows from \$5,023.14 to \$6,112.14 at a 3.7% annual interest rate compounded continuously, how much time passed during this growth?

$$A = Pe^{rt}$$

$$6,112.14 = 5,023.14e^{0.037t}$$

$$\frac{6,112.14}{5,023.14} = \frac{5,023.14e^{0.037t}}{5,023.14}$$

$$1.21679... = e^{0.037t}$$

$$0.037t = \log_e 1.21679...$$

$$0.037t = \ln 1.21679...$$

$$0.037t = 0.19622...$$

$$\boxed{t \approx 5.3 \text{ years}}$$

- ③③ If an \$800.99 investment grows to \$31,024 in 30 years, what interest rate was applied if it was compounded continuously?

$$A = Pe^{rt}$$

$$31,024 = 800.99e^{r(30)}$$

$$31,024 = 800.99e^{30r}$$

$$\frac{31,024}{800.99} = \frac{800.99e^{30r}}{800.99}$$

$$38.732... = e^{30r}$$

$$30r = \log_e 38.732...$$

$$30r = \ln 38.732...$$

$$30r = 3.656...$$

$$r = 0.121888...$$

$$\boxed{12.19\%}$$

- ③④ If an investment grows from \$126,476 to \$130,000 at a 4% annual interest rate compounded continuously, how much time passed during this growth?

$$A = Pe^{rt}$$

$$130,000 = 126,476e^{0.04t}$$

$$\frac{130,000}{126,476} = \frac{126,476e^{0.04t}}{126,476}$$

$$1.02786... = e^{0.04t}$$

$$0.04t = \log_e 1.02786...$$

$$0.04t = \ln 1.02786...$$

$$0.04t = 0.02748...$$

$$\boxed{t \approx 0.687 \text{ years}}$$

The Change of Base Formula

Logarithm Property IX

$$\log_A B = \frac{\log_K B}{\log_K A}$$

* Notice that "K" appears on the right side of the equation but not on the left. This means that "K" can be any number that you choose (as long as it is positive).

Use Logarithm Property IX to change each logarithm to a quotient of logarithms with the indicated base.

③⑤ Base 5.

$$\log_6 9 = \frac{\log_5 9}{\log_5 6}$$

③⑥ Base 3.

$$\log_8 7 = \frac{\log_3 7}{\log_3 8}$$

③⑦ Base 9.

$$\log_4 5 = \frac{\log_9 5}{\log_9 4}$$

③⑧ Base 49.

$$\log_{98} 45 = \frac{\log_{49} 45}{\log_{49} 98}$$

• ③⑨ Base 2.

$$\log_7 12 = \frac{\log_2 12}{\log_2 7}$$

④⑩ Base 4.

$$\log_5 3 = \frac{\log_4 3}{\log_4 5}$$

④① Base 6.

$$\log_6 100 = \boxed{\frac{\log_6 100}{\log_6 6}}$$

④② Base 8.

$$\log_8 17 = \boxed{\frac{\log_8 17}{\log_8 8}}$$

Using Calculators with Only Base "e" and Base 10 Capability

Use your calculator's base 10 logarithm button (LOG) and Logarithm Property IX to find the value of each logarithm below. Round to the nearest thousandth.

$$\bullet \text{ ④③ } \log_3 7 = \frac{\log_{10} 7}{\log_{10} 3} = \frac{\log 7}{\log 3} \approx \boxed{1.7712}$$

$$\text{④④ } \log_5 2 = \frac{\log 2}{\log 5} \approx \boxed{0.4307}$$

$$\bullet \text{ ④⑤ } \log_9 32 = \frac{\log 32}{\log 9} \approx \boxed{1.5773}$$

$$\text{④⑥ } \log_8 6 = \frac{\log 6}{\log 8} \approx \boxed{0.8617}$$

Use your calculator's base "e" logarithm button (LN) and Logarithm Property IX to find the value of each logarithm below. Round to the nearest thousandth.

$$\bullet \text{ ④⑦ } \log_4 7 = \frac{\log_e 7}{\log_e 4} = \frac{\ln 7}{\ln 4} \approx \boxed{1.4}$$

$$\text{④⑧ } \log_6 2 = \frac{\ln 2}{\ln 6} \approx \boxed{0.39}$$

$$\bullet \text{ ④⑨ } \log_8 11 = \frac{\ln 11}{\ln 8} \approx \boxed{1.15}$$

$$\text{⑤⑩ } \log_{15} 17 = \frac{\ln 17}{\ln 15} \approx \boxed{1.05}$$